

		Rescue Junior - Bill of Materials (BOM)				Name of Local Currency: Canadian Dollar				
						Total Cost (Local Currency):		\$1,834.41		
		Team Name:	Kavosh_Line			Total Cost (USA Dollars):		\$1,336.91		
	#	Component	Part Name	Author	Source	Quantity	Unit Cost (Local)	Unit Cost (USA)	Total Cost (Local)	Total Cost (USA)
	1	Drive Motor	Dynamixel XC430-T240BB-T	Dynamixel	robotis.us	4	\$192.98	\$137.89	\$771.94	\$551.56
	2	Arm Motor	Dyanmixel AX-12A	Dynamixel	robotis.us	1	\$80.32	\$57.39	\$80.32	\$57.39
	3	Claw and Back Motor	Dynamixel XL-320	Dynamixel	robotis.us	3	\$55.90	\$30.94	\$129.91	\$92.82
	4	Custom PCBs	Custom Made	Carson & Ethan / Jlcpcb	jlcpcb.com	2	\$50.00	\$35.73	\$300.00	\$214.35
	5	RGB Sensors	APDS9960	Adafruit	leeselectronic.com	3	\$18.00	\$12.86	\$54.00	\$38.58
	6	Distance Sensors	VL53L1X ToF Distance Sensor	Pololu	pololu.com	4	\$32.12	\$22.95	\$91.80	\$91.80
	7	Distance Sensor	VL53L7CX ToF FOV Distance Sensor	Pololu	pololu.com	1	\$27.92	\$19.95	\$27.92	\$19.95
	8	Voltage Step Down	3.3V 2.5A Step-Down Pololu	Pololu	robotshop.com	1	\$45.89	\$32.79	\$45.89	\$32.79
	9	Distance IR Sensor	GP2Y0A21YK0F	Sharp	leeselectronic.com	1	\$20.00	\$14.29	\$20.00	\$14.29
	10	3D Printing Filament	PLA Filament 1.75mm Black & Red	ELEGOO	amazon.ca	2	\$20.39	\$14.57	\$40.78	\$29.14
	11	Connecting Parts	Screw, Nuts, and Washers		leeselectronic.com	1	\$30.00	\$21.44	\$30.00	\$21.44
	12	Black Line Sensors	TCRT5000	Visha Semiconductors	mouser.ca	16	\$1.48	\$1.06	\$23.68	\$16.92
	13	Battery	Zeee 3S Lipo Battery 11.1V 2200mAH	Zeee	amazon.ca	1	\$21.67	\$15.48	\$21.67	\$15.48
	14	Electronical Components	Wires, Cables, Resistors, Soldering Tools, Display, Switch		leeselectronic.com	1	\$150.00	\$107.18	\$150.00	\$107.18
	15	Microcontroller	TEENSY 4.1	TEENSY	leeselectronic.com	1	\$46.50	\$33.22	\$46.50	\$33.22

(shipment cost)

		Rescue Junior - Bill of Materials (BOM)				Name of Local Currency: Canadian Dollar		
		Team Name:	Kavosh_Line			Total Cost (Local Currency):		\$41.98
						Total Cost (USA Dollars):		\$30.00
#	Name	Software's Tool / Library	Description	Author	Source	Cost (Local)	Cost (USA)	
1	Dynamixel2Arduino	Library	Controls Dynamixel X-series servos, reads position, velocity, and current temperature	ROBOTIS	GitHub	Free	Free	
2	Adafruit APDS9960 Library	Library	For sensing light and colour using the APDS9960	Adafruit	GitHub	Free	Free	
3	Pololu VL53L1X Library	Library	For sensing precise distance in millimetres using the VL53L1X ToF Distance Sensor	Pololu	GitHub	Free	Free	
4	STM VL53L7CX Library	Library	For sensing a 8x8 distance map using the VL53L7CX ToF FOV Distance Sensor	STMicroelectronics	GitHub	Free	Free	
5	Adafruit SSD1306	Library	For displaying outputs from sensors using an OLED display	Adafruit Industries	GitHub	Free	Free	
6	Teensyduino Core Libraries	Library	Controls the Teensy 4.1 Microcontroller	Teensy	pjrc.com	Free	Free	
7	Arduino IDE	Software's Tool	Software for writing, compiling, and uploading code to the robot	Arduino	arduino.com	Free	Free	
8	Fusion 360	Software's Tool	CAD of the robot and 3D printing parts	Autodesk	autodesk.com	Free	Free	
9	Solidworks	Software's Tool	CAD of the robot and 3D printing parts	Dassault Systemes	solidworks.com	\$41.98	\$30.00	(2 month subscription)
10	Altium Designer	Software's Tool	Designing custom PCBs	Altium Ltd.	altium.com	Free	Free	(borrowed from a parent)
11	RoboPlus	Software's Tool	Changing Dynamixel motor settings	ROBOTIS	emanual.robotis.com	Free	Free	